Example 7.7 From a solid cylinder whose height is 2.4 cm and diameter 1.4 cm, a conical cavity of the same height and base is hollowed out (Fig.7.13). Find the total surface area of the remaining solid.

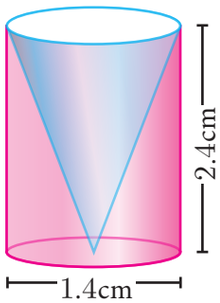


Fig. 7.13

Solution Let h and r be the height and radius of the cone and cylinder.

Let l be the slant height of the cone.

Given that, $h = 2.4$ cm and $d = 1.4$ cm ; $r = 0.7$ cm

$$\left.\begin{matrix}\text{Total surface area of the}\\ \text{remaining solid}\end{matrix}\right\} = \begin{matrix}\text{C.S.A. of the cylinder + C.S.A. of the cone}\\ \text{+ area of the bottom}\end{matrix}$$

$$= 2\pi rh + \pi rl + \pi r^2 \text{ sq. units}$$

$$= \pi r\left(2h + l + r\right) \text{ sq. units}$$

Now, $$l = \sqrt{r^2 + h^2} = \sqrt{0.49 + 5.76} = \sqrt{6.25} = 2.5 \text{ cm}$$

$$l = 2.5 \text{ cm}$$

$$\text{Area of the remaining solid} = \pi r\left(2h + l + r\right) \text{ sq. units}$$

$$= \frac{22}{7} \times 0.7 \times \left[\left(2 \times 2.4\right) + 2.5 + 0.7\right]$$

$$= 17.6$$

Therefore, total surface area of the remaining solid is 17.6 cm^2

Progress Check

1. Right circular cone is a solid obtained by revolving _____ about _____.
2. In a right circular cone the axis is _____ to the diameter.
3. The difference between the C.S.A. and T.S.A. of a cone is _____.
4. When a sector of a circle is transformed to form a cone, then match the conversions taking place between the sector and the cone.

Sector	Cone
Radius	Circumference of the base
Area	Slant height
Arc length	Curved surface area

7.2.4 The Sphere

Definition : A sphere is a solid generated by the revolution of a semicircle about its diameter as axis.

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